TP-Link N300 WiFi Router (TL-WR841N)

Visit the manufacturer's website devoted to this router at
https://www.tp-link.com/ca/home-networking/wifi-router/tl-wr841n/

Use the information at this site to answer the following questions and perform any of the specified exercises:

1. At what speed does this router operate wirelessly?
   2.4GZ
2. How many LAN ports does it have?
   4 LAN ports
3. What is connected to the LAN ports?
   Ethernet Cables
4. How many WAN ports does it have?
   1 WAN port
5. What is connected to the WAN port?
   The ISP cable
6. Is it compatible with IPv6?
   Yes
7. What is IPv6 and how does it compare to IPv4?
   A new protocol for ip addresses that use 128 bits of hexadecimal digits instead of 32 bits of decimal digits
8. What level of encryption can it use?
   WPA2
9. What wireless transmission standard does it use?
   802. 11n
10. Does it support WPS?
    yes
11. What is WPS and how is it useful in a wireless router?
    It means wifi protected set up and allows you to quickly connect to the router
12. What wireless wavelength does it operate on?
    2.4ghz
13. How fast are its hard wired ports?
    10 mtps
14. Does it support Static and Dynamic WAN IP assignments?
    Yes
15. Does it support DHCP?
    Yes
16. There are three lights on the front of this router. Explain them.
    The lights are power, if its connected to the world wide internet, and if theres a wfi signal

17. Explain all the parts you see in the following image of the back of the router:



Blue = WAN cable that connects to ISP cable
Orange = LAN cables that connect devices within LAN
Button on the right = reset button that resets all passwords and previous settings
Port on the left = power cable
Wireless antennas

18. What url can you use to log into the router's web interface?
    You can use the routers IP address, which is 192.168.0.1
19. Once you find the URL please connect your computer to the LAN port of the router and log into the user interface. Once you have logged in, go to the status page. What is the WAN IP and LAN IP address of the router?
    LAN: 192.160.0.1
    WAN: n/a (no isp)
20. Open the Windows Console window or Linux Terminal Window and ping the LAN IP Address. What happens and why?
    It sends packets of data to the router and sees how fast they take to get sent back, it's used to test connecton
21. What is the subnet mask of the LAN interface? Why is this important to know?
    The subnet mask is 255.255.255.0

    It's important because it shows you what digits of the IP address represent the network address and what digits are unique to the devices on your network

22. Open a new tab in your browser and log into the router using the LAN IP address.

23. Go back to the status page. What is the MAC address of the LAN and WAN interfaces?
    LAN: 40:ED:00:DF:EF:08
    WAN: 40:ED:00:DF:EF:09

24. What is a MAC address and why are these different?
    A 12 digit hexadecimal digit that uniquely identifies devices within a network. They are different because the LAN and WAN interface is different hardware each with their own MAC Addresses

25. What is the MAC address of the wireless interface? What is the wireless SSID?
    MACL: 40:ED:00:DF:EF:08

    SSID: TP-link-EF08

26. Click on the DHCP tab? What is the purpose of the Start and End IP Addresses text boxes?
    They are 192.168.0.100(start) and 192.168.0.199(end) they show the range of IP addresses that the router can assign

27. What are the Primary and Secondary DNS IP Addresses used for?

28. Click on the DHCP→DHCP Client List. What would you see here and why?
    I see my computer and it's information(ip and mac address) because it's connected to the router

29. Plug in another computer to this router and refresh this page. What happens and why?

There is another computer on the client list, listing it's ip and mac address. This happens because the computer is connected to the router along with our computer

30. Click on the Parental Control and explain the role of the settings on this page.
    Parental controls administrator activities on the network and limit/prevent access to certain websites on all client connected to the network

31. Explain the System Tools→Firmware Upgrade tab page.
32. Go to System Tools→Password. Change the settings so that only you and your group can log in to the router. Restart the router and open a new web page and attempt to log in to the router using the new settings.
33. Now that you have a second computer connected to your router, use the router to find the IP addresses of all connected PCs. Enter the windows console of each computer and enter the command **ipconfig**. Explain the information that you see.
34. Try pinging each computer from another computer connected to the network (make sure to turn off all firewalls in each computer first) . Embed a screenshot of your windows console results.
    It sends a broadcast message to all computers on the network causing them to attempt to reply

35. From either computer perform a ping test such that the address you are pinging ends in 255. What are the results you get? Explain them.
36. Once you have successfully pinged all computers on your LAN, download a LAN game and have some fun!

Look at the image below and take note of all the devices required for you to have the internet in your home:

